

Cost-Sensitive Predictive Modeling for Targeted Marketing Campaigns

PROJECT 2 · ADVANCED MACHINE LEARNING

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Outline

Problem and Data

Baseline Methods

Proposed Pipeline

Experiments

Results and Conclusions

Problem and Data

Business Setup and Scoring Rule

SCENARIO

A bank selects up to 1 000 customers from 5 000 to contact with a product offer.

Converts	+10 EUR
Ignores	−5 EUR
Per variable	−200 EUR

BREAK-EVEN THRESHOLD

$$\mathbb{E}[\text{gain} \mid \hat{p}] = 15\hat{p} - 5 > 0, \text{ so } \hat{p} > \frac{1}{3}$$

SCORING FORMULA

$$S = 10 \cdot \text{TP} - 5 \cdot \text{FP} - 200 \cdot K$$

Each extra variable must generate >200 EUR in added profit. Accuracy and profit are decoupled objectives.

Dataset and EDA Highlights

DATASET

- 5 000 train + 5 000 test observations
- 500 anonymised features (V1–V500)
- Nearly balanced: 49.76% positives
- No missing or constant features

KEY EDA FINDINGS

- Weak feature–target correlations ($\rho_{\max} = 0.114$, V390)
- 23 pairs with $|\rho| > 0.80$
- Two perfect collinearities: V32 $\equiv$ V175, V160 $\equiv$ V416
- Including both: $\times 2$ cost, 0 gain

Multivariate model needed, but 200 EUR per variable forces aggressive, profit-aware selection.

Baseline Methods

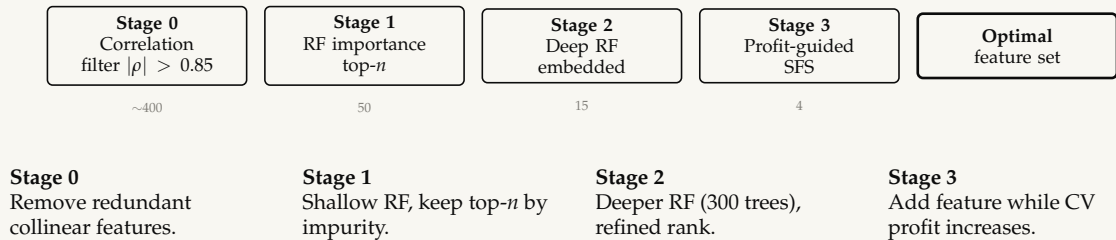
Why Standard Feature Selection Fails

Approach	K	Data cost	Val. acc.	Profit
LR, single feature (V390)	1	200 EUR	0.538	+2 280 EUR
LR, top-5 correlated	5	1 000 EUR	0.578	+1 490 EUR
LR, Boruta features (26)	26	5 200 EUR	0.605	−2 730 EUR
HGB, Boruta features (26)	26	5 200 EUR	0.680	−2 730 EUR
LR, L1 regularisation	56	11 200 EUR	0.565	−8 750 EUR
Proposed pipeline	4	800 EUR	—	+5 165 EUR (CV)

HGB on Boruta features achieves 68% accuracy (the highest baseline) yet yields −2 730 EUR. The 5 200 EUR data cost overwhelms the gain. Higher accuracy $\neq$ higher profit.

Proposed Pipeline

Four-Stage Profit-Driven Feature Selector



Stage 3: Greedy SFS and Threshold Optimisation

GREEDY STOPPING CRITERION

Marginal gain of candidate f at step t :

$$\Delta_t(f) = \hat{S}_{CV}(\mathcal{F}_t \cup \{f\}) - \hat{S}_{CV}(\mathcal{F}_t) - 200$$

Add $\arg \max_f \Delta_t(f)$ if positive, else stop.

AFTER FEATURE SELECTION

- 24-iter randomised HGB search
- HGB-only vs. ensemble (HGB+RF+LR) compared by OOF profit
- Threshold grid $[0.15, 0.50]$, 36 points

WHY NOT FILTER / EMBEDDED ALONE?

Filter uses surrogate importance scores. SFS evaluates the actual financial objective with a real classifier, including the 200 EUR marginal cost.

THRESHOLD OPTIMISATION

$$\tau^* = \arg \max_{\tau} [10 \text{TP}(\tau) - 5 \text{FP}(\tau)]$$

Variable cost 200K is τ -independent.

Experiments

Experiment: Free Feature Engineering

MOTIVATION

Having paid for $K = 6$ variables, generate additional features at zero extra cost.

Transformation	Formula	Count
Pairwise products	$f_i \cdot f_j$	$\binom{6}{2} = 15$
Directed ratios	$f_i / (f_j + \varepsilon)$	30
Signed log	$\text{sign}(f) \log(1 + f)$	6
Row statistics	min, max, mean, std, range	5
PCA + K-means	2 components + 5 cluster features	7
Total		63 derived features

Result: No architecture improved CV profit on the 69-feature matrix. The same six original variables were re-selected. *The original features already capture the available signal.*

Feature Selection Configuration Search

Three configurations of Stages 0–2, evaluated by 5-fold CV on all 5 000 training samples:

Configuration	Filter n	Embedded m	Corr. thr.	K	CV Profit
Standard	55	15	0.85	4	5 165 EUR
Wide net	100	30	0.90	3	5 035 EUR
Strict net	40	10	0.85	4	5 165 EUR

Standard and Strict converge to the same four features:

V160 V191 V215 V32

Identical result across independent configurations validates the SFS stopping criterion.

All Tested Pipelines

Pipeline	K	Profit	Context
LR, top-1 feature	1	+2 280 EUR (val)	acc. 0.54
LR, top-5 features	5	+1 490 EUR (val)	acc. 0.58
Boruta + HGB	26	−2 730 EUR (val)	acc. 0.68, cost 5 200 EUR
LR, L1	56	−8 750 EUR (val)	highest data cost
RF-only (tuned)	4	<5 165 EUR (CV)	
Calibrated HGB + thr. search	4	<5 165 EUR (CV)	
Free feature engineering	4	≈5 020 EUR (CV)	no improvement
Main pipeline	4	5 165 EUR (CV)	winner

All variants select the same six features. HGB outperforms RF-only consistently.

Results & Conclusions

Final Model Performance

SELECTED FEATURE SET ($K = 4$)

V160 V191 V215 V32

- $< 0.8\%$ of the original 500 features
- Data cost: $4 \times 200 = 800$ EUR
- 1 000 customers targeted (full budget)
- Decision threshold: $\tau^* = 0.170$

PROFIT ESTIMATES

Estimate	Value
OOF CV profit (biased)	5 165 EUR
Nested CV profit (unbiased)	$315 \pm 1\,162$ EUR
Projected test score	$\approx 5\,900$ EUR

Unbiased estimate from 5-fold nested CV where feature selection runs inside each fold. Wide interval reflects small validation folds (1 000 samples each).

Expected Test Score and Conclusions

PROJECTED TEST SCORE

At 78% precision on 1 000 targets:

$$780 \times 10 - 220 \times 5 - 4 \times 200 = \mathbf{5,900 \text{ EUR}}$$

Approach	Profit
L1 / Boruta	−8 750 to −2 730 EUR
Best baseline (LR, 1 var)	+2 280 EUR
Proposed	≈+5 900 EUR

CONCLUSIONS

1. **Objective > model.** Profit-guided SFS is the decisive design choice.
2. **Stable solution.** Two independent configs + 5 alternative pipelines all converge to the same 4 features.
3. **Free features add nothing.** 63 derived features gave no CV gain.
4. **More vars ≠ better.** 56-var L1 scores −8 750 EUR; 4 vars score +5 900 EUR.

Thank you

Questions?

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